

INTEGER SEQUENCES / GENERATING FUNCTIONS / ELLIPTIC CURVES

Elliptic addition and Jacobi continued fractions

An explicit proof of a conjectured correspondence,
with indexing and torsion corrections

Research report prepared with ChatGPT

19 September 2026

Abstract

Paul Barry's 2023 paper *Integer sequences from elliptic curves* conjectures that the Jacobi continued-fraction coefficients of a particular algebraic generating function are given by coordinates of successive multiples of a point on an elliptic curve. We give an explicit algebraic proof of a consistently indexed, non-torsion formulation of that assertion. The main construction is a closed expression for every normalized complete quotient. A factorization shows that deleting one continued-fraction layer is exactly translation by the distinguished elliptic-curve point, after a rational change of coordinates. No inference from finitely many coefficients is used in the proof.

The corresponding formulas for both coordinates in terms of Hankel and modified Hankel determinants follow from a self-contained moment argument. We also derive the associated Somos-4 recurrence and recover the already-known division-polynomial identity, explicitly crediting the 2026 proof of Liu, Wang, and Zhang. Torsion specializations explain an essential missing hypothesis: a zero Hankel determinant need not cause all later ones to vanish. In fact, the generating function is nonrational whenever the parameter b is nonzero, even in these exceptional cases. Exact computations on 106 nonsingular curves, sixteen symbolic identities, and four worked examples accompany the article. The literature audit identifies the precise conjecture addressed and distinguishes the present proof from previously established results; priority is not asserted.

Central correspondence. On $Y^2 + aXY + bY = X^3 + cX^2 + dX$, write $mP = (X_m, Y_m)$ for $P = (0, 0)$. For the generating function defined in Section 2, the regular Jacobi coefficients are

$$\alpha_0 = 1, \quad \alpha_1 = -1, \quad \beta_1 = 1, \quad \alpha_m = \frac{bY_m}{X_m} - d - 1, \quad \beta_m = -b^2X_m \quad (m \geq 2).$$

The proof constructs all the tails, rather than merely guessing these coefficients.

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1 The problem and the literature boundary

1.1 The published conjecture

Barry associates an integer-coefficient generating function to a generalized Weierstrass equation with a specified integral point. In Section 3 of [1], Conjecture 2 concerns its Hankel transform and an elliptic division sequence; Conjecture 3 expresses point coordinates through Hankel determinants; Conjecture 4 proposes a Jacobi continued fraction in the reverse direction. We take the last assertion as the main target, with the coordinate formulas as its natural consequence.

This is a tractable kind of sequence problem because the generating function has degree two over a rational-function field. Its continued-fraction tails can plausibly remain in the same quadratic extension. The difficulty is not computing a few tails: it is finding a closed family stable under the continued-fraction transformation and proving that its parameter evolution agrees with the elliptic group law.

1.2 What was already known by the research date

The current literature matters here. Liu, Wang, and Zhang’s preprint [2], submitted in August 2026, explicitly lists Barry’s 2023 Conjecture 2 among the assertions it proves. Its Theorems 5.4–5.5 give the Somos-4 recurrence and the normalized Hankel/division-polynomial identity. Those are therefore *not* claimed as new results in this report.

The general relationship among continued fractions, elliptic translation, orthogonal polynomials, and Hankel determinants is older. Hone’s work [3] provides a broad treatment, including higher genus. Barry’s 2025 paper [4] develops related lattice-path and continued-fraction correspondences, and Chang and Liu [5] treat further issues involving hyperelliptic functions and bilateral Somos sequences. The present calculation belongs to this established framework; it is not a discovery of the general connection.

The specific deliverable here is a directly checkable complete-quotient formula in the four parameters of [1], together with a proof that it yields the intended assertions of Conjectures 3–4. The sources inspected did not supply this exact proof in this normalization. That is a bounded literature observation, not a guarantee that no equivalent result or unpublished proof exists. Thus the report proves a precise version of a published conjectural statement, but does not certify a first-in-the-literature resolution of a still-open problem as of September 2026.

1.3 Two necessary corrections

There are two different issues, which should not be conflated.

First, the sentence immediately before Conjecture 4 in [1] identifies (x_n, y_n) with $n(0, 0)$, while its displayed formulas use $\beta_{n+1} = -b^2x_n$ and $\alpha_n = by_{n-1}/x_{n-1} - d - 1$. Taken literally, the first nontrivial coefficient would involve $x_1 = 0$, and α_2 would contain $0/0$. The same paper’s coordinate formulas initialize $(x_0, y_0) = (0, 0)$. The consistent interpretation is

$$(x_n, y_n) = ((n + 1)P), \quad n \geq 0, \tag{1.1}$$

not nP . We instead write $(X_m, Y_m) = mP$ throughout, reserving the point at infinity O for $0P$. This removes the ambiguity.

Second, a point can be torsion. Then some multiples reach O or $-P = (0, -b)$, and the displayed affine ratios cease to exist. An infinite *regular* Jacobi fraction consequently requires a nondegeneracy hypothesis. Infinite order of P supplies exactly what is needed here. Section 8 explains what remains valid at torsion

points. A typographical indexing contradiction alone would be an uninteresting “counterexample”; our goal is to prove the substantive intended formula.

2 Normalization and the main theorem

Let K be a field of characteristic zero. Consider the nonsingular curve

$$E : Y^2 + aXY + bY = X^3 + cX^2 + dX, \quad P = (0, 0), \quad b \neq 0. \quad (2.1)$$

The coefficients a, b, c, d lie in K . Write $mP = (X_m, Y_m)$ whenever $mP \neq O$. Introduce

$$\delta = abd - b^2c + d^2, \quad \kappa = b^4, \quad (2.2)$$

$$s = ab + 2d + 2, \quad r = s - 1, \quad L = \delta + r. \quad (2.3)$$

In particular,

$$L = (d + 1)^2 + ab(d + 1) - b^2c.$$

The letter δ does not denote the discriminant of E .

Define the quadratic polynomial and quartic polynomial

$$M(z) = 1 + sz + Lz^2, \quad \mathcal{D}(z) = M(z)^2 - 4\kappa z^3(1 + rz). \quad (2.4)$$

There is a unique $W(z) \in K[[z]]$ with

$$W(z)^2 = \mathcal{D}(z), \quad W(0) = 1. \quad (2.5)$$

Comparing the first four coefficients gives

$$W(z) = 1 + sz + Lz^2 - 2\kappa z^3 + O(z^4). \quad (2.6)$$

The series $q(z)$ and the moment generating function $G(z)$ are

$$q(z) = \frac{M(z) - W(z)}{2\kappa z^3}, \quad (2.7)$$

$$G(z) = \frac{1}{1 - z - z^2 q(z)} = \frac{2\kappa z}{W(z) + 2\kappa z(1 - z) - M(z)}. \quad (2.8)$$

The apparent powers of z in the denominators cancel. In particular, $q(0) = G(0) = 1$. Equivalently, q is the unique solution with constant term one of

$$M(z)q(z) - \kappa z^3 q(z)^2 = 1 + rz. \quad (2.9)$$

This compact normalization is also compatible with the quadratic-series form used in [2, Section 5.2]. Appendix A checks it against Barry’s expanded coefficients.

Definition 2.1. A regular Jacobi continued fraction is a formal expansion

$$\frac{1}{1 - \alpha_0 z - \frac{\beta_1 z^2}{1 - \alpha_1 z - \frac{\beta_2 z^2}{1 - \alpha_2 z - \dots}}}, \quad \beta_j \neq 0 \quad (j \geq 1). \quad (2.10)$$

Its value is the coefficientwise, or z -adic, limit of its finite truncations. This definition involves no analytic convergence assertion.

Theorem 2.2 (Corrected elliptic Jacobi-fraction formula). *Assume that P has infinite order on (2.1). The generating function (2.8) has the regular Jacobi expansion (2.10) with*

$$\alpha_0 = 1, \quad \alpha_1 = -1, \quad \beta_1 = 1, \quad (2.11)$$

and, for every $m \geq 2$,

$$\alpha_m = \frac{bY_m}{X_m} - d - 1, \quad \beta_m = -b^2 X_m. \quad (2.12)$$

These formulas are the consistently indexed version of Barry's Conjecture 4. The expansion is unique among regular Jacobi fractions.

The proof is given in Sections 3–5. Its core is an elementary factorization, not a determinant calculation.

2.1 Why all denominators exist

If P has infinite order, then $mP \neq O$ for every positive integer m . The only points on (2.1) with $X = 0$ are $P = (0, 0)$ and $-P = (0, -b)$. If mP were either of these for $m \geq 2$, then $(m-1)P = O$ or $(m+1)P = O$, a contradiction. Consequently

$$X_m \neq 0 \quad (m \geq 2). \quad (2.13)$$

In particular $\beta_m \neq 0$. This simple observation is the exact place where the infinite-order hypothesis enters the infinite construction.

2.2 Integer coefficients without elliptic divisions

Write

$$q(z) = \sum_{n \geq 0} q_n z^n, \quad G(z) = \sum_{n \geq 0} \mu_n z^n.$$

Equation (2.9) gives $q_0 = 1$, $q_1 = -1$, and for $n \geq 2$,

$$q_n = -s q_{n-1} - L q_{n-2} + \kappa \sum_{i=0}^{n-3} q_i q_{n-3-i}, \quad (2.14)$$

where a sum with negative upper limit is zero. The outer fraction gives $\mu_0 = 1$ and

$$\mu_n = \mu_{n-1} + \sum_{i=0}^{n-2} q_i \mu_{n-2-i} \quad (n \geq 1). \quad (2.15)$$

Thus all q_n and μ_n belong to $\mathbb{Z}[a, b, c, d]$. The rational elliptic coefficients in (2.12) produce polynomial moments; this is a consequence of the algebraic series identity, not an assumption about cancellation.

For example,

$$\begin{aligned} (q_0, q_1, q_2, q_3) &= (1, -1, 1 - \delta, \delta(s+1) - 1 + \kappa), \\ (\mu_0, \dots, \mu_5) &= (1, 1, 2, 2, 4 - \delta, 4 + \kappa + (s-1)\delta). \end{aligned}$$

The recurrences remain well-defined when the Jacobi formula degenerates.

3 Elliptic addition as a rational map

3.1 Coordinates adapted to the fraction

For a point $Q = (X, Y)$ with $X \neq 0$, set

$$u = \frac{bY}{X} - d - 1, \quad v = -b^2X. \quad (3.1)$$

The inverse transformation is

$$X = -\frac{v}{b^2}, \quad Y = -\frac{v(u + d + 1)}{b^3}. \quad (3.2)$$

Lemma 3.1 (Invariant cubic). *The point Q lies on E if and only if*

$$\mathcal{I}(u, v) := v^2 + (u^2 + su + L)v - \kappa(u + 1) = 0. \quad (3.3)$$

Here the equivalence is on the open set $v \neq 0$.

Proof. Put $T = u + d + 1$. Substitution of (3.2) into the left side minus the right side of (2.1), followed by multiplication by b^6/v , gives

$$v^2 + v(T^2 + abT - b^2c) - b^4(T - d).$$

The identities

$$T^2 + abT - b^2c = u^2 + su + L, \quad T - d = u + 1$$

turn this expression into (3.3). $\square$

3.2 The translation law

The line through $P = (0, 0)$ and $Q = (X, Y)$ has slope $t = Y/X$. Substitution of $Y = tX$ into (2.1) shows that the three intersection abscissas, with multiplicity, satisfy

$$X(X^2 + (c - t^2 - at)X + (d - bt)) = 0.$$

The group law negates the third intersection point. Hence, if $Q + P = (X', Y')$,

$$X' = t^2 + at - c - X, \quad Y' = -(t + a)X' - b. \quad (3.4)$$

This formula includes tangent intersections when multiplicities occur.

Lemma 3.2 (Conjugate translation map). *Under (3.1), translation $Q \mapsto Q + P$ becomes*

$$v' = -v - u^2 - su - L = -\frac{\kappa(u + 1)}{v}, \quad (3.5)$$

$$u' = \frac{\kappa}{v'} - s - u, \quad (3.6)$$

whenever v and v' are nonzero.

Proof. Multiply the first formula in (3.4) by $-b^2$ and use $bt = u + d + 1$. This yields the first expression for v' in (3.5). The second follows by dividing (3.3) by v .

For the ordinate,

$$\frac{bY'}{X'} - d - 1 = -b(t + a) - \frac{b^2}{X'} - d - 1 = -u - s + \frac{\kappa}{v'},$$

which is (3.6). Invariance of (3.3) follows already from the group law; it can also be checked by direct substitution. $\square$

Thus the geometric orbit provides a sequence

$$(u_m, v_m) = (\alpha_m, \beta_m), \quad m \geq 2,$$

satisfying

$$v_{m+1} + v_m = -(u_m^2 + su_m + L), \quad (3.7)$$

$$u_{m+1} + u_m = \frac{\kappa}{v_{m+1}} - s. \quad (3.8)$$

The next section shows that these same relations govern the tails of G .

3.3 The starting point is $2P$

Implicit differentiation of (2.1) at P gives tangent slope d/b . Consequently

$$X_2 = \frac{\delta}{b^2}, \quad Y_2 = -\left(\frac{d}{b} + a\right)X_2 - b. \quad (3.9)$$

Under the infinite-order hypothesis, $\delta \neq 0$ by (2.13). Substitution into (3.1) gives

$$v_2 = -\delta, \quad u_2 = -r - \frac{\kappa}{\delta}. \quad (3.10)$$

This initialization is another direct check of the required index shift.

4 Explicit complete quotients

Definition 4.1. For a pair (u, v) satisfying (3.3), with $v \neq 0$, define

$$F_{u,v}(z) = \frac{1 + sz + (L + 2v)z^2 - W(z)}{2vz^2(1 + (\kappa/v - u)z)}. \quad (4.1)$$

This is a normalized complete quotient, meaning a tail with constant term one.

By (2.6), its numerator begins $2vz^2 + 2\kappa z^3$. The apparent division by z^2 is therefore harmless and

$$F_{u,v}(z) = 1 + uz + O(z^2). \quad (4.2)$$

Lemma 4.2 (The factorization). *Let $N_v(z) = 1 + sz + (L + 2v)z^2$ and $\gamma = \kappa/v - u$. On the invariant cubic,*

$$N_v(z)^2 - W(z)^2 = 4vz^2(1 + \gamma z)(1 + (s + u)z). \quad (4.3)$$

In particular,

$$\frac{1}{F_{u,v}(z)} = \frac{N_v(z) + W(z)}{2(1 + (s + u)z)}. \quad (4.4)$$

Proof. Since $N_v = M + 2vz^2$, subtraction using (2.4) gives

$$\begin{aligned} N_v^2 - W^2 &= 4vz^2M + 4v^2z^4 + 4\kappa z^3(1 + rz) \\ &= 4vz^2 \left(1 + \left(s + \frac{\kappa}{v}\right)z + \left(L + v + \frac{\kappa r}{v}\right)z^2 \right). \end{aligned}$$

The two proposed linear factors have the correct linear coefficient. Their quadratic coefficient is correct precisely when

$$\left(\frac{\kappa}{v} - u\right)(s + u) = L + v + \frac{\kappa(s - 1)}{v}.$$

After multiplication by v , this equality is equivalent to (3.3). This proves (4.3). Multiplying numerator and denominator of the reciprocal of (4.1) by $N_v + W$ now proves (4.4). All cancelled factors are nonzero formal series or explicit powers of z . $\square$

Theorem 4.3 (One translation deletes one layer). *Suppose (u, v) satisfies (3.3), and (u, v) and (u', v') are related by (3.5)–(3.6), with $v, v' \neq 0$. Then*

$$F_{u,v}(z) = \frac{1}{1 - uz - v'z^2 F_{u',v'}(z)}. \quad (4.5)$$

Proof. Equation (3.6) gives

$$\frac{\kappa}{v'} - u' = s + u.$$

Thus the denominator of $v'z^2 F_{u',v'}$ is $2(1 + (s + u)z)$. By (4.4), the numerator of $1 - uz - 1/F_{u,v}$ over that same denominator is

$$\begin{aligned} & 2(1 - uz)(1 + (s + u)z) - N_v(z) - W(z) \\ &= 1 + sz + (-2u(s + u) - L - 2v)z^2 - W(z) \\ &= 1 + sz + (L + 2v')z^2 - W(z). \end{aligned}$$

The last equality is exactly (3.5). This is the numerator of $v'z^2 F_{u',v'}$, proving the identity. $\square$

Remark 4.4. The construction is local and algebraic. The proof of theorem 4.3 needs the invariant cubic, the displayed rational map, and nonzero denominators; it does not require an independent theorem about elliptic functions or a prior Somos recurrence. Nonsingularity and infinite order are used to identify and continue the geometric orbit globally.

5 Proof of the continued-fraction theorem

5.1 The initial two layers

By rationalizing (2.7),

$$\frac{1}{q(z)} = \frac{M(z) + W(z)}{2(1 + rz)}. \quad (5.1)$$

For the initial pair (3.10), one has

$$\frac{\kappa}{v_2} - u_2 = r.$$

Hence (4.1) specializes to

$$F_{u_2,v_2}(z) = \frac{W(z) - 1 - sz + (\delta - r)z^2}{2\delta z^2(1 + rz)}. \quad (5.2)$$

Direct substitution into (5.1) gives

$$q(z) = \frac{1}{1 + z + \delta z^2 F_{u_2,v_2}(z)} = \frac{1}{1 - \alpha_1 z - \beta_2 z^2 F_{u_2,v_2}(z)}. \quad (5.3)$$

The definition of G supplies the outer layer,

$$G(z) = \frac{1}{1 - \alpha_0 z - \beta_1 z^2 q(z)}, \quad \alpha_0 = 1, \quad \beta_1 = 1. \quad (5.4)$$

This proves all the special initial values and links the algebraic series to the first geometric tail.

5.2 Iteration and formal convergence

For every $m \geq 2$, put $F_m = F_{u_m, v_m}$, where (u_m, v_m) is obtained from mP . Lemma 3.2 and Theorem 4.3 give

$$F_m = \frac{1}{1 - \alpha_m z - \beta_{m+1} z^2 F_{m+1}}. \quad (5.5)$$

All tails exist by (2.13). Iterating (5.5) inside (5.3) and (5.4) gives the proposed fraction.

For completeness, define

$$T_{A,B}(H) = \frac{1}{1 - Az - Bz^2 H}.$$

If H and $\tilde{H}$ are formal series with constant term one, then

$$T_{A,B}(H) - T_{A,B}(\tilde{H}) = \frac{Bz^2(H - \tilde{H})}{(1 - Az - Bz^2 H)(1 - Az - Bz^2 \tilde{H})}. \quad (5.6)$$

Each denominator is a unit. Therefore each surrounding layer increases an error's order by at least two. Replacing F_m by 1 makes an $O(z)$ error at that level, hence an $O(z^{2m+1})$ error in G . The finite expressions stabilize coefficientwise, and their limit is exactly G . This proves existence in theorem 2.2.

5.3 Uniqueness

If $F = 1 + f_1 z + f_2 z^2 + \dots$ is a normalized tail, then any regular representation $F = (1 - Az - Bz^2 H)^{-1}$ forces

$$A = -[z]F^{-1}, \quad B = -[z^2]F^{-1}, \quad H = \frac{1 - Az - F^{-1}}{Bz^2}.$$

Thus the coefficient pair and next tail are uniquely determined whenever $B \neq 0$. Induction proves uniqueness of the entire regular fraction. Together with the preceding subsections, this completes the proof of theorem 2.2.

6 Hankel determinants and both point coordinates

Let $\mu_n = [z^n]G(z)$. Define

$$h_n = \det(\mu_{i+j})_{0 \leq i, j \leq n}, \quad h_{-1} = 1. \quad (6.1)$$

The modified determinant h_n^* has the same first n rows as h_n , but its last row is shifted one place:

$$h_n^* = \det A^{(n)}, \quad A_{ij}^{(n)} = \begin{cases} \mu_{i+j}, & i < n, \\ \mu_{n+j+1}, & i = n, \end{cases} \quad 0 \leq i, j \leq n. \quad (6.2)$$

Set $h_{-1}^* = 0$. In particular $h_0 = 1$ and $h_0^* = \mu_1 = 1$. The star does not mean that every entry in the matrix has been shifted; that different convention would give a different determinant.

6.1 The formal moment lemma

Lemma 6.1 (Jacobi coefficients from determinants). *For a regular fraction (2.10), with moment sequence μ_n ,*

$$h_n = \prod_{j=1}^n \beta_j^{n+1-j}, \quad (6.3)$$

$$\beta_n = \frac{h_n h_{n-2}}{h_{n-1}^2} \quad (n \geq 1), \quad (6.4)$$

$$\alpha_n = \frac{h_n^*}{h_n} - \frac{h_{n-1}^*}{h_{n-1}} \quad (n \geq 0). \quad (6.5)$$

For $n = 1$, formula (6.4) uses $h_{-1} = 1$.

Proof. We give a formal proof, so positivity and analytic moment measures are unnecessary. Let V be the vector space with basis $e_0, e_1, \dots$, and let J be the tridiagonal operator

$$J e_n = e_{n+1} + \alpha_n e_n + \beta_n e_{n-1}, \quad \beta_0 e_{-1} := 0.$$

The coefficient of e_0 in $J^k e_0$ is the total weight of all length- k paths on the nonnegative integers starting and ending at zero. An up-step has weight one, a horizontal step at height n has weight α_n , and a down-step from height n has weight β_n . Decomposing paths at their first horizontal step or first up-and-down excursion gives (2.10). Consequently that coefficient is μ_k .

Set $w_0 = 1$ and $w_n = \beta_1 \cdots \beta_n$. Give V the symmetric bilinear form $\langle e_i, e_j \rangle = w_i \mathbf{1}_{i=j}$. Then J is self-adjoint because $w_{n+1} = \beta_{n+1} w_n$. Let

$$\mathcal{L}(f) = \langle f(J) e_0, e_0 \rangle; \quad \mathcal{L}(t^k) = \mu_k.$$

Define monic polynomials by $p_0 = 1$, $p_1 = t - \alpha_0$, and

$$p_{n+1} = (t - \alpha_n) p_n - \beta_n p_{n-1}.$$

Induction gives $p_n(J) e_0 = e_n$. Self-adjointness therefore implies

$$\mathcal{L}(p_i p_j) = w_i \mathbf{1}_{i=j}.$$

The basis change from $1, t, \dots, t^n$ to $p_0, \dots, p_n$ is unitriangular. The Gram determinant in the monomial basis is h_n , whereas in the polynomial basis it is $\prod_{j=0}^n w_j$. This proves (6.3), and its adjacent ratios give (6.4).

Finally, the coefficient of t^n in p_{n+1} is $-\sum_{j=0}^n \alpha_j$ by the three-term recurrence. The determinant formula for the monic polynomial orthogonal to $1, t, \dots, t^n$ is

$$p_{n+1}(t) = \frac{1}{h_n} \begin{vmatrix} \mu_0 & \mu_1 & \cdots & \mu_{n+1} \\ \mu_1 & \mu_2 & \cdots & \mu_{n+2} \\ \vdots & \vdots & & \vdots \\ \mu_n & \mu_{n+1} & \cdots & \mu_{2n+1} \\ 1 & t & \cdots & t^{n+1} \end{vmatrix}.$$

Indeed it is monic, and applying $\mathcal{L}(t^j \cdot)$ for $j \leq n$ creates two equal rows. Expansion along the last row shows that its t^n coefficient is $-h_n^*/h_n$: the relevant minor is the transpose of (6.2). Thus $h_n^*/h_n = \sum_{j=0}^n \alpha_j$. Subtracting the formula at $n-1$ proves (6.5). $\square$

6.2 The coordinate theorem

Corollary 6.2 (Corrected form of Barry's Conjecture 3). *Under the hypotheses of theorem 2.2, all h_n are nonzero. For $m \geq 2$,*

$$X_m = -\frac{1}{b^2} \frac{h_{m-2}h_m}{h_{m-1}^2}, \quad (6.6)$$

$$Y_m = -\frac{1}{b^3} \frac{h_{m-2}h_m}{h_{m-1}^2} \left(\frac{h_m^*}{h_m} - \frac{h_{m-1}^*}{h_{m-1}} + d + 1 \right). \quad (6.7)$$

Together with $(X_1, Y_1) = (0, 0)$, these recover all positive multiples of P .

Proof. Nonvanishing follows from (6.3). Combine (2.12) with (6.4) and (6.5), and then solve the first equation in (3.1) for Y_m . $\square$

Replacing m by $n + 1$ gives the intended formulas in [1, Conjecture 3]. The star determinants are necessary to recover the ordinate with its sign; the unmodified Hankel ratios alone recover only the abscissa.

7 The known Somos and division-polynomial consequences

This section gives an independent route from the complete-quotient proof to results already established in [2, Theorems 5.4–5.5]. It is included to connect the proof to integer sequences and to control exceptional specializations, not to claim a new Somos-4 theorem.

7.1 A second-order recurrence for the Jacobi numerators

Proposition 7.1. *For the nondegenerate orbit, the numerator coefficients satisfy*

$$v_{m-1}v_m^2v_{m+1} = \kappa^2(v_m + \delta), \quad m \geq 3. \quad (7.1)$$

The lower bound $m \geq 3$ is essential: the exceptional value $\beta_1 = 1$ is not a member of the geometric orbit beginning at $2P$.

Proof. Write $u = u_m$, $v = v_m$, and use (3.8) at $m - 1$:

$$u_{m-1} = \frac{\kappa}{v} - s - u.$$

Since $v_{m-1}v = -\kappa(u_{m-1} + 1)$,

$$v_{m-1} = \frac{\kappa(s + u - 1)}{v} - \frac{\kappa^2}{v^2}, \quad v_{m+1} = -\frac{\kappa(u + 1)}{v}.$$

Multiplication gives

$$v_{m-1}v^2v_{m+1} = \kappa^2 \left(-(u + 1)(s + u - 1) + \frac{\kappa(u + 1)}{v} \right).$$

By (3.3), the last fraction is $v + u^2 + su + L$. The bracket simplifies to $v + L - s + 1 = v + \delta$. $\square$

7.2 The bilinear recurrence and its normalization

Using (6.4), the left side of (7.1), with $m = n - 1$, is

$$\frac{h_n h_{n-4}}{h_{n-2}^2}.$$

Therefore, for $n \geq 4$,

$$h_n h_{n-4} = b^8 h_{n-1} h_{n-3} + b^8 \delta h_{n-2}^2. \quad (7.2)$$

Define

$$S_n = b^{-n^2+2n} h_n. \quad (7.3)$$

The differences between the quadratic exponents on the two sides are six and eight, respectively. Hence

$$S_n S_{n-4} = b^2 S_{n-1} S_{n-3} + \delta S_{n-2}^2. \quad (7.4)$$

The first four values are

$$S_0 = 1, \quad S_1 = b, \quad S_2 = -\delta, \quad S_3 = -b((s-2)\delta + \kappa). \quad (7.5)$$

Indeed $h_0 = h_1 = 1$, $h_2 = v_2 = -\delta$, and $h_3 = v_2^2 v_3 = -\kappa((s-2)\delta + \kappa)$ by (3.10) and (3.5).

7.3 Identification with division polynomials

Let ψ_n be the standard division polynomials of (2.1), and put $D_n = \psi_n(P)$. The initial values needed here are

$$D_0 = 0, \quad D_1 = 1, \quad D_2 = b, \quad D_3 = -\delta, \quad D_4 = -b((s-2)\delta + \kappa). \quad (7.6)$$

For example, in the standard Weierstrass invariants, $b_4 = ab + 2d = s - 2$, $b_6 = b^2$, and $b_8 = -\delta$. Thus $D_3 = b_8$ and $D_4 = b(b_4 b_8 - b_6^2)$. The classical division-polynomial addition identity, recorded in [2, Equation (5.5)], yields

$$D_{j+2} D_{j-2} = b^2 D_{j+1} D_{j-1} + \delta D_j^2. \quad (7.7)$$

Corollary 7.2 (Known identity, recovered here). *For every $n \geq 0$,*

$$b^{-n^2+2n} h_n = \psi_{n+1}(P). \quad (7.8)$$

The identity holds for arbitrary characteristic-zero specializations with $b \neq 0$; nonsingularity is required only when interpreting it as a statement about the elliptic group law.

Proof. First work over the generic field $\mathbb{Q}(a, b, c, d)$. The generic point P is non-torsion: a hypothetical generic identity $NP = \mathcal{O}$ would force $\psi_N(P)$ to be the zero polynomial, contrary to its specialization at $E : Y^2 + Y = X^3 - X$, whose point $(0, 0)$ has infinite order as shown in Section 9.1. The standard characterization $\psi_j(P) = 0$ if and only if $jP = \mathcal{O}$ is used here.

All terms are therefore nonzero in the generic field. Equations (7.4) and (7.7), with the matching initial values (7.5)–(7.6), identify S_n with D_{n+1} by induction. This proves (7.8) generically.

Both sides belong to $\mathbb{Q}[a, b, b^{-1}, c, d]$: the moments and their Hankel determinants are polynomials by (2.14)–(2.15), and division polynomials are polynomials in the Weierstrass coefficients and point coordinates. Equality in the fraction field is consequently an identity in this ring. It specializes at every $b \neq 0$, even when some later D_j or h_j vanish. No division by a specialized zero is performed in this argument. $\square$

The proof above also explains why the division-free form (7.2) specializes safely, whereas solving it for h_n by dividing by h_{n-4} need not do so. This distinction is important at torsion points.

8 Torsion, vanishing determinants, and nonrationality

8.1 A complete description of the zero locations

Corollary 8.1. *Assume E is nonsingular and $b \neq 0$. If P has finite order N , then $N \geq 3$ and*

$$h_n = 0 \iff N \mid n + 1. \quad (8.1)$$

Thus the first zero is h_{N-1} , but the next determinant h_N is nonzero. If P has infinite order, no h_n vanishes.

Proof. Since $-P = (0, -b) \neq P$, the point P is not of order two, and it is not O . Apply (7.8) and the vanishing characterization of division polynomials. $\square$

At order N , the point $(N-1)P = -P$ has abscissa zero. Formally the numerator value $-b^2 X_{N-1}$ is therefore zero, but the ordinate ratio bY_{N-1}/X_{N-1} is undefined. There is no valid ordinary Jacobi tail with these specialized scalar coefficients. It would be incorrect to interpret the zero numerator by itself as a terminating fraction.

8.2 Why the fraction cannot terminate

Proposition 8.2. *For every characteristic-zero parameter choice with $b \neq 0$, the series $q(z)$ and $G(z)$ in (2.7)–(2.8) are nonrational over $K(z)$. This conclusion does not require nonsingularity or infinite order.*

Proof. If $W(z)$ were rational, the equality $W^2 = \mathcal{D} \in K[z]$ would force $W \in K[z]$: in a reduced fraction, a nonconstant denominator cannot have its square divide the square of the numerator. Since $\deg \mathcal{D} \leq 4$, one would then have $\deg W \leq 2$.

But (2.6) has cubic coefficient $-2\kappa = -2b^4 \neq 0$. This is impossible. Thus W is not rational. Equation (2.7) shows that rationality of q is equivalent to rationality of W . Equation (2.8) shows that rationality of G would imply that of $q = (1 - z - G^{-1})/z^2$. Both series are therefore nonrational. $\square$

A finite ordinary Jacobi fraction is rational. Hence theorem 8.2 gives a second, independent reason why torsion specializations cannot be handled by naive termination. Equivalently, for $b \neq 0$ the moment sequence cannot satisfy any eventual homogeneous linear recurrence with constant coefficients in K : multiplying a rational generating function by its denominator gives such a recurrence, and conversely such a recurrence makes the generating function rational.

8.3 Two explicit torsion examples

For

$$E_3 : Y^2 + Y = X^3, \quad (8.2)$$

we have $(a, b, c, d) = (0, 1, 0, 0)$, $\delta = 0$, and curve discriminant $-27 \neq 0$. The tangent at P is $Y = 0$, which meets the cubic three times at P . Thus $3P = O$. The moment and Hankel sequences begin

$$\begin{aligned} (\mu_n) &= (1, 1, 2, 2, 4, 5, 6, 15, 6, 35, 25, 17, 185, \dots), \\ (h_n) &= (1, 1, 0, -1, -1, 0, 1, 1, 0, \dots). \end{aligned}$$

Here $h_2 = 0$ but $h_3 = -1$. If one set $\beta_2 = 0$ and truncated after $\alpha_1 = -1$, one would obtain

$$\frac{1}{1 - z - z^2/(1 + z)} = \frac{1 + z}{1 - 2z^2},$$

whose coefficient of z^5 is 4, while the actual coefficient is $\mu_5 = 5$. The failed specialization is visible at a very small index.

For

$$E_4 : Y^2 + XY + Y = X^3 + X^2, \quad (8.3)$$

we have $2P = (-1, 0)$, which is a nonidentity point of order two. Thus P has order four. The curve discriminant is -15 and

$$(h_n) = (1, 1, 1, 0, -1, -1, -1, 0, 1, \dots).$$

Again the isolated zero does not propagate to every later determinant. These examples refute the unqualified extension of the ordinary nondegenerate fraction to all nonsingular curves; they do not refute theorem 2.2.

9 Worked non-torsion examples

9.1 A small example and a certificate of infinite order

Take

$$E : Y^2 + Y = X^3 - X, \quad P = (0, 0). \quad (9.1)$$

Then $\delta = \kappa = 1$, $s = 0$, $r = -1$, $L = 0$, and

$$W(z) = \sqrt{1 - 4z^3 + 4z^4}, \quad q(z) = \frac{1 - W(z)}{2z^3}, \quad G(z) = \frac{2z}{W(z) + 2z - 2z^2 - 1}. \quad (9.2)$$

The discriminant is 37, so the curve is nonsingular.

There are elementary certificates that P has infinite order. For a direct certificate using computed point coordinates, the change of variables

$$U = 4X, \quad V = 8Y + 4$$

transforms (9.1) into $V^2 = U^3 - 16U + 16$. The first few additions give $7P = (-5/9, 8/27)$, so its transformed abscissa is $-20/9$. The Nagell–Lutz theorem says that rational torsion points on an integral short Weierstrass equation have integral coordinates; see [6, Theorem 24.21]. Therefore $7P$, and hence P , has infinite order. This uses a theorem with explicit hypotheses, not an inference from a long list of nonzero multiples.

The same conclusion can be checked through good reduction: $\#E(\mathbb{F}_3) = 7$ and $\#E(\mathbb{F}_5) = 8$. The rational torsion subgroup injects into these good reductions, so its order divides both numbers and is one; see [6, Corollary 24.19]. The verification script counts these finite sets directly.

The moments and determinants begin

$$\begin{aligned} (\mu_n) &= (1, 1, 2, 2, 3, 4, 4, 6, 7, 6, 11, 10, 6, \dots), \\ (h_n) &= (1, 1, -1, 1, 2, -1, -3, -5, 7, -4, -23, \dots), \\ (h_n^*) &= (1, 0, 0, 1, -1, 3, 2, 6, -13, -5, 45, \dots). \end{aligned}$$

For example, the coordinate formulas at $m = 4$ give

$$X_4 = -\frac{h_2 h_4}{h_3^2} = 2, \quad Y_4 = X_4 \left(\frac{h_4^*}{h_4} - \frac{h_3^*}{h_3} \right) = 2 \left(-\frac{1}{2} - 1 \right) = -3.$$

Here $d + 1 = 0$, so there is no additional constant in the ordinate bracket.

m	X_m	Y_m	α_m	β_m
2	1	0	0	-1
3	-1	-1	1	1
4	2	-3	-3/2	-2
5	1/4	-5/8	-5/2	-1/4
6	6	14	7/3	-6
7	-5/9	8/27	-8/15	5/9
8	21/25	-69/125	-23/35	-21/25

Table 1: Exact elliptic coordinates and continued-fraction coefficients for $Y^2 + Y = X^3 - X$. The special outer values are $\alpha_0 = 1$, $\alpha_1 = -1$, and $\beta_1 = 1$.

9.2 Barry's larger numerical example

Barry's motivating example [1, Section 2] uses

$$(a, b, c, d) = (2, 5, 4, 9), \quad E : Y^2 + 2XY + 5Y = X^3 + 4X^2 + 9X.$$

The discriminant is -38091 , and direct counting gives $\#E(\mathbb{F}_5) = 8$ and $\#E(\mathbb{F}_7) = 6$. Both reductions are good, so the rational torsion subgroup has order dividing two. Since $-P = (0, -5) \neq P$, the point P cannot be torsion. Our normalization gives

$$\delta = 71, \quad \kappa = 625, \quad s = 30, \quad r = 29, \quad L = 100,$$

and hence

$$G(z) = \frac{1250z}{\sqrt{1 + 60z + 1100z^2 + 3500z^3 - 62500z^4 + 1220z - 1350z^2 - 1}}. \quad (9.3)$$

The initial elliptic values are

$$2P = \left(\frac{71}{25}, -\frac{1974}{125} \right), \quad \alpha_2 = -\frac{2684}{71}, \quad \beta_2 = -71.$$

These agree with direct stripping of the moment series. The next point is

$$3P = \left(\frac{65325}{5041}, \frac{14724605}{357911} \right),$$

which gives

$$\alpha_3 = \frac{1089691}{185523}, \quad \beta_3 = -\frac{1633125}{5041}.$$

The relatively complicated rational coefficients nevertheless generate the integer moment sequence

$$1, 1, 2, 2, -67, 2688, -73696, 1856194, -44867225, 1067747230, \dots$$

Its normalized Hankel sequence is

$$1, 5, -71, -13065, -1275214, 2876558965, 11607013366079, \dots$$

The example is included as a normalization check against the original paper; all the displayed values are also recomputed independently in the accompanying files. The main proof is not based on this example.

10 Exact verification and reproducibility

10.1 Independent computational routes

The file `verify.py` uses Python’s standard library only. Its calculations use arbitrary-precision integers and exact fractions; no floating-point tests or numerical tolerances occur. The following routes are deliberately separate.

The moment series is generated from the division-free quadratic recurrences (2.14)–(2.15). Elliptic multiples are computed with the generalized Weierstrass chord-and-tangent formulas, not by the rational map (3.5)–(3.6). Jacobi coefficients are extracted from the moment series by formal reciprocal and coefficient stripping, not by substituting elliptic coordinates. Hankel and modified Hankel determinants are computed by fraction-free Bareiss elimination with row pivoting. Division-polynomial values are computed with the classical odd-index and doubling recurrences, whose only scalar divisor after evaluation at P is $D_2 = b$. They remain usable when later terms vanish.

The default exhaustive parameter set is

$$a, c, d \in \{-1, 0, 1\}, \quad b \in \{-2, -1, 1, 2\}. \quad (10.1)$$

It contains 108 curves; the two singular ones are explicitly skipped for the group-law tests. For the remaining 106 curves, the script checks all nondegenerate layers up to $m = 8$, for a total of 544 layer checks. It checks (7.8) for every $0 \leq n \leq 9$ on each of the 106 curves, including torsion cases.

The four named examples are additionally exported through μ_{40} , h_{12} , h_{12}^* , and $12P$. Their division-polynomial identities are checked through $n = 12$. The finite Catalan coefficient formula in Appendix B is independently checked for $0 \leq n \leq 20$ in all four examples, giving 84 finite-sum checks. All these tests passed in the delivered run.

10.2 Symbolic checks and their scope

The optional file `verify_symbolic.py`, tested with SymPy 1.14.0, verifies sixteen rational-function or polynomial identities. These include the conversion of Barry’s quartic coefficients, the birational invariant cubic, the complete-quotient factorization, the one-step numerator identity, the initial tail identity, and the Somos relation for the numerator coefficients. The program reduces identities on the invariant curve by substituting

$$L = \frac{\kappa(u+1)}{v} - v - u^2 - su,$$

which is valid on the same open set $v \neq 0$ used in the proof. Every symbolic check passed.

The numerical and symbolic tests are audits of the written argument, not proof substitutes. They cannot establish novelty, check every future specialization by enumeration, or replace the nonvanishing and formal-limit arguments in Sections 2 and 5. No Lean or other proof-assistant formalization is claimed.

10.3 Running the package

From the extracted archive’s main directory, run

```
python verify.py --output data
python verify_symbolic.py      # optional: requires sympy
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command regenerates the exact data and a machine-readable verification summary. The optional symbolic audit does not affect the standard-library verification. The two LaTeX runs resolve the contents, cross-references, and bibliography. A Makefile supplies the same actions. The bibliography is embedded in `article.tex`, so a BibTeX run or network connection is not required.

The quadratic recurrences need $O(N^2)$ arithmetic operations for the first N moments using direct convolution. Computing all determinants individually through index N with cubic-time elimination uses $O(N^4)$ arithmetic operations. These are arithmetic-operation counts, not bit-complexity claims: integer and rational numerators can become large. The code prioritizes transparent independent verification rather than a new fastest algorithm.

11 Conclusion: what has been established

The main result is a proof of the naturally corrected, non-torsion version of the elliptic Jacobi-fraction formula stated as Conjecture 4 in [1]. Its decisive ingredient is the explicit tail $F_{u,v}$ in (4.1). The invariant equation of the elliptic curve factors its algebraic norm; that factorization makes continued-fraction stripping identical to elliptic translation. The argument proves all coefficients at once and supplies the formal convergence justification.

The Hankel formulas recover both coordinates, giving the correspondingly indexed form of Conjecture 3. The same calculation recovers the known Somos-4 and division-polynomial statements, while the universal polynomial identity explains exactly what survives at torsion specializations. Such specializations expose a genuine domain issue: ordinary regular Jacobi fractions break down, but the algebraic generating function remains defined and nonrational, and Hankel determinants resume being nonzero after an isolated zero.

The mathematical claim and the historical claim have different scopes. The complete algebraic proof and its tests are provided here. The broader elliptic/continued-fraction theory and the normalized Hankel identity are already part of the literature. Establishing bibliographic priority for this particular explicit proof would require further specialist review; it is not inferred merely from the word “conjecture” in an older paper.

A Matching the original expanded generating function

Barry’s generating function is written as

$$g(z) = \frac{2b^4z}{\sqrt{Az^4 + Bz^3 + Cz^2 + D_1z + 1 + Fz^2 + G_1z - 1}}.$$

The present parameters give the much shorter coefficient list

$$A = L^2 - 4kr, \quad B = 2sL - 4\kappa, \quad (A.1)$$

$$C = s^2 + 2L, \quad D_1 = 2s, \quad (A.2)$$

$$F = -(L + 2\kappa), \quad G_1 = 2\kappa - s. \quad (A.3)$$

For example,

$$\begin{aligned} L^2 - 4kr &= a^2b^2(d+1)^2 - 2ab(2b^4 + b^2c(d+1) - (d+1)^3) \\ &\quad + b^4(c^2 - 4(2d+1)) - 2b^2c(d+1)^2 + (d+1)^4, \end{aligned}$$

which is Barry’s coefficient A . The other identities follow by direct expansion and are included in the symbolic audit. Thus the proof concerns the original generating function, not merely another series sharing a few initial coefficients.

B Finite coefficient formulas from Catalan numbers

Let

$$C(t) = \sum_{k \geq 0} C_k t^k = \frac{1 - \sqrt{1 - 4t}}{2t}, \quad C_k = \frac{1}{k+1} \binom{2k}{k}.$$

Factoring M out of (2.7) gives the formal identity

$$q(z) = \frac{1 + rz}{M(z)} C\left(\frac{\kappa z^3(1 + rz)}{M(z)^2}\right) = \sum_{k \geq 0} C_k \kappa^k z^{3k} (1 + rz)^{k+1} M(z)^{-2k-1}. \quad (\text{B.1})$$

No convergence condition is needed: the argument of C has order three. To extract a completely finite expression, put

$$T_{k,m}(s, L) = \sum_{\ell=0}^{\lfloor m/2 \rfloor} (-1)^{m-\ell} \binom{2k+m-\ell}{m-\ell} \binom{m-\ell}{\ell} s^{m-2\ell} L^\ell. \quad (\text{B.2})$$

Then

$$q_n = \sum_{k=0}^{\lfloor n/3 \rfloor} C_k \kappa^k \sum_{j=0}^{\min(k+1, n-3k)} \binom{k+1}{j} r^j T_{k, n-3k-j}(s, L). \quad (\text{B.3})$$

To verify (B.2), expand $(1 + sz + Lz^2)^{-2k-1}$ using the negative binomial theorem. If ℓ factors contribute Lz^2 , the total number of nonconstant factors is $m - \ell$, and the remaining $m - 2\ell$ contribute sz . This gives exactly the displayed summand. Equation (B.3) now follows from (B.1).

There is also a finite composition expression for each moment. From (2.8),

$$G(z) = \sum_{j \geq 0} \frac{z^{2j} q(z)^j}{(1 - z)^{j+1}}.$$

For $n \geq 0$, this implies

$$\mu_n = 1 + \sum_{j=1}^{\lfloor n/2 \rfloor} \sum_{\substack{k_1, \dots, k_j \geq 0 \\ k_1 + \dots + k_j \leq n-2j}} \binom{n-j-(k_1 + \dots + k_j)}{j} \prod_{i=1}^j q_{k_i}. \quad (\text{B.4})$$

Substituting (B.3) removes the auxiliary sequence entirely. These formulas provide recurrence-free finite sums for the moments, but (2.14)–(2.15) are usually more convenient for computation.

C Verification details and boundary conventions

The standard-library verifier uses the following division-polynomial recurrences after evaluation at P :

$$\begin{aligned} D_{2m+1} &= D_{m+2} D_m^3 - D_{m-1} D_{m+1}^3 \quad (m \geq 2), \\ b D_{2m} &= D_m (D_{m-1}^2 D_{m+2} - D_{m-2} D_{m+1}^2) \quad (m \geq 3), \end{aligned}$$

with the base values (7.6). These are distinct from the Somos recurrence used in the proof and do not divide by a possibly zero D_{m-2} . Consequently they provide a useful independent test in the torsion cases.

The curve discriminant used to exclude singular inputs is

$$\Delta_E = -b_2^2 b_8 - 8b_4^3 - 27b_6^2 + 9b_2 b_4 b_6,$$

where

$$b_2 = a^2 + 4c, \quad b_4 = ab + 2d, \quad b_6 = b^2, \quad b_8 = -\delta.$$

There is no extra factor of 1728 in this formula. The parameters Δ_E and δ are kept separate in the code.

In the point-data CSV files, the row $m = 0$ is explicitly the point at infinity. Rows with $X_m = 0$ mark the coordinate ratio as undefined; the special outer Jacobi values α_0 , α_1 , and β_1 are not inferred from those rows. A stopping point in series stripping is reported only as a failure of regular scalar stripping at that depth, not as a claim that the full generating function terminates.

The archive includes the source, compiled article, verification programs, logs, JSON summaries, and exact CSV tables. It does not include external paper PDFs, installed fonts, or generated LaTeX auxiliary files.

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